

How to READ the Notation

Speaking Math Out Loud

1 The Problem

Mathematical notation looks like random symbols until you know how to read it. Let's fix that!

Goal: By the end of this guide, you'll be able to read any expression in this problem OUT LOUD in plain English.

2 Basic Symbols - How to Say Them

Symbol	How to Say It	What It Means
N_t	"N sub t" or "N at time t"	Number of arrivals by time t
N_s	"N sub s" or "N at time s"	Number of arrivals by time s
N_4	"N sub 4" or "N at time 4"	Number of arrivals by time 4
N_9	"N sub 9" or "N at time 9"	Number of arrivals by time 9
N_{14}	"N sub 14" or "N at time 14"	Number of arrivals by time 14
λ	"lambda"	The rate (arrivals per unit time)
n	"n"	A specific number (like 13)

3 Reading Conditions - The Vertical Bar

The vertical bar $|$ means "given that" or "if we know that".

Symbol	How to Say It
$(N_4 N_9 = 13)$	"N-four given N-nine equals 13" OR "N at time 4, given that N at time 9 equals 13"
$(N_t N_s = n)$	"N-t given N-s equals n" OR "N at time t, given that N at time s equals n"
$P(N_4 = 5 N_9 = 13)$	"The probability that N-four equals 5, given N-nine equals 13"

4 Reading Complete Expressions

Let's read some actual expressions from the homework:

4.1 Example 1: $N_9 = 13$

Mathematical: $N_9 = 13$

Read as: “N sub nine equals thirteen”

Plain English: “There are 13 arrivals by time 9” or “By time 9, 13 customers have arrived”

4.2 Example 2: $P(N_4 = 5 \mid N_9 = 13)$

Mathematical: $P(N_4 = 5 \mid N_9 = 13)$

Read as: “The probability that N-four equals five, given that N-nine equals thirteen”

Plain English: “What’s the probability there were 5 arrivals by time 4, if we know there were 13 arrivals by time 9?”

4.3 Example 3: $(N_t \mid N_s = n)$

Mathematical: $(N_t \mid N_s = n)$

Read as: “N-t, given N-s equals n”

Plain English: “The number of arrivals by time t, if we know there were n arrivals by time s”

4.4 Example 4: $N_9 - N_4 = 8$

Mathematical: $N_9 - N_4 = 8$

Read as: “N-nine minus N-four equals eight”

Plain English: “The number of arrivals between time 4 and time 9 equals 8”

Think: If 13 arrived by time 9, and 5 arrived by time 4, then $13 - 5 = 8$ arrived between times 4 and 9.

4.5 Example 5: $(N_t \mid N_s = n) \sim \text{Binomial}(n, \frac{t}{s})$

Mathematical: $(N_t \mid N_s = n) \sim \text{Binomial}(n, \frac{t}{s})$

Read as: “N-t given N-s equals n, follows a Binomial distribution with parameters n and t-over-s”

Plain English: “If we know there were n arrivals by time s, then the number of arrivals by the earlier time t follows a Binomial distribution where we do n trials, each with probability t/s”

4.6 Example 6: $N_t = n + \text{Poisson}(\lambda(t - s))$

Mathematical: $N_t = n + \text{Poisson}(\lambda(t - s))$

Read as: “N-t equals n plus a Poisson random variable with parameter lambda times quantity t-minus-s”

Plain English: “The number of arrivals by time t equals the n we already know about, plus a random number following Poisson with rate lambda times the time difference (t minus s)”

5 Common Confusing Notation

5.1 Subscripts vs Variables

Important Distinction:

- N_4 means a SPECIFIC time: time 4 (4 hours, 4am, etc.)
- N_t means a VARIABLE time: time t (could be anything)
- N_s means another VARIABLE time: time s (different from t)

When you see N_t and N_s , think: “time t” and “time s” where t and s are just placeholders for any two times.

When you see N_4 and N_9 , think: specific times 4 and 9.

5.2 What Does $\sim$ Mean?

The symbol $\sim$ means “follows the distribution” or “is distributed as”.

Symbol	How to Say It
$X \sim \text{Binomial}(n, p)$	“X follows a Binomial distribution with parameters n and p”
$Y \sim \text{Poisson}(\lambda)$	“Y follows a Poisson distribution with parameter lambda”

6 Reading Question 4b

The Question: For any $t, s \geq 0$, what is the conditional distribution of N_t given $N_s = n$?

Read as: “For any times t and s (both greater than or equal to zero), what is the conditional distribution of N-t, given that N-s equals n?”

Plain English: “Pick any two times, call them t and s. If I tell you that n arrivals happened by time s, what can you tell me about the distribution of arrivals by time t?”

7 Step-by-Step Reading Practice

Let’s practice reading the answer to 4b:

7.1 Case 1 Answer

Written: $(N_t \mid N_s = n) \sim \text{Binomial}\left(n, \frac{t}{s}\right)$ when $t < s$

Read it slowly:

1. “N-t” - the number of arrivals by time t
2. “given” - if we know that
3. “N-s equals n” - there were n arrivals by time s
4. “follows” - has the distribution

5. “Binomial” - Binomial distribution
6. “with n trials” - we do n trials
7. “and probability t-over-s” - each with success probability t divided by s
8. “when t is less than s” - and this is when time t comes before time s

Put it together: “The number of arrivals by time t, given that there were n arrivals by time s, follows a Binomial distribution with n trials and probability t-over-s, when t is less than s.”

7.2 Case 3 Answer

Written: $(N_t \mid N_s = n) = n + \text{Poisson}(\lambda(t - s))$ when $t > s$

Read it slowly:

1. “N-t” - the number of arrivals by time t
2. “given” - if we know that
3. “N-s equals n” - there were n arrivals by time s
4. “equals” - is equal to
5. “n plus” - n (the known arrivals) plus
6. “Poisson lambda times t-minus-s” - a Poisson random variable with parameter lambda times the quantity t minus s
7. “when t is greater than s” - when time t comes after time s

Put it together: “The number of arrivals by time t, given there were n arrivals by time s, equals n plus a Poisson random variable with rate lambda times t-minus-s, when t is greater than s.”

8 The Actual Coffee Shop Example

Let’s read the concrete example with numbers:

8.1 Example: $(N_4 \mid N_9 = 13) \sim \text{Binomial}(13, \frac{4}{9})$

Symbol by symbol:

- N_4 = “N at time 4” = “arrivals by 4am”
- $\mid$ = “given that”
- $N_9 = 13$ = “N at time 9 equals 13” = “13 arrivals by 9am”
- $\sim$ = “follows the distribution”
- $\text{Binomial}(13, \frac{4}{9})$ = “Binomial with 13 trials, probability 4/9”

Full reading: “The number of arrivals by 4am, given that there were 13 arrivals by 9am, follows a Binomial distribution with 13 trials and success probability 4/9.”

Even simpler: “If 13 customers arrived by 9am, the number who arrived by 4am is like flipping 13 coins, each with a 4/9 chance of heads.”

9 Practice Problems

Try reading these yourself, then check the answers:

9.1 Problem 1

Symbol: $N_{14} = 28$

Your answer: _____

Correct answer: “N at time 14 equals 28” OR “There are 28 arrivals by time 14”

9.2 Problem 2

Symbol: $P(N_t = k \mid N_s = n)$

Your answer: _____

Correct answer: “The probability that N-t equals k, given N-s equals n” OR “The probability of k arrivals by time t, given n arrivals by time s”

9.3 Problem 3

Symbol: $(N_{10} \mid N_5 = 8) = 8 + \text{Poisson}(10)$

Your answer: _____

Correct answer: “N at time 10, given N at time 5 equals 8, equals 8 plus Poisson with parameter 10” OR “Arrivals by time 10, given 8 arrivals by time 5, equals 8 plus a Poisson random variable with rate 10”

10 Quick Reference Card

Notation Cheat Sheet

- N_t = “number of arrivals by time t”
- $N_s = n$ = “n arrivals happened by time s”
- $\mid$ = “given that” or “if we know”
- $\sim$ = “follows the distribution” or “is distributed as”
- $(N_t \mid N_s = n)$ = “arrivals by time t, given n arrivals by time s”
- $N_t - N_s$ = “arrivals BETWEEN time s and time t”
- λ = “lambda” = the rate
- $\lambda(t - s)$ = “lambda times t-minus-s” = rate times time interval

11 Tips for Reading Math

1. **Go slow:** Read symbol by symbol at first
2. **Say it out loud:** Hearing yourself helps

3. **Translate to English:** After reading symbols, say what it means
4. **Use concrete numbers:** Replace variables with actual numbers
5. **Draw it:** Sketch a timeline if it helps

Remember: The notation is just a shorthand. Every symbol has a plain English meaning. Once you know how to “translate,” the math becomes much easier!